

4(a)	<i>four from:</i> <i>vasodilation</i> 1 arterioles / blood vessels, (in skin) widen ; R capillaries I arteries 2 more blood flows to capillaries / blood flows through capillaries more slowly ; 3 heat (energy) lost ; <i>sweating</i> 4 more sweat, produced / released ; 5 sweat / water, evaporates ; 6 using heat energy / <i>ref. to</i> latent heat of vaporisation ;	4
4(b)	<i>four from:</i> 1 high blood glucose concentration / hyperglycaemia ; 2 (causes) decrease in water potential (of blood) ; 3 detected by, osmoreceptors / hypothalamus ; 4 feelings of thirst ; 5 less / no, glucose converted to, fat / glycogen ; 6 glucose lost in urine / not all glucose reabsorbed / blood glucose concentration above the renal threshold ; 7 glucose not taken up by cells ; 8 (so) fats / proteins, are, respired / metabolised ;	4

Question	Answer	Marks
4(c)(i)	<i>three from:</i> 1 (pad contains) glucose oxidase ; 2 enzyme / glucose oxidase, reacts with glucose (in the blood) ; 3 oxygen detected ; 4 (electric) current generated ; 5 detected by electrode ; 6 gives numerical value (of blood glucose concentration) ;	3
4(c)(ii)	<i>one from:</i> 1 gives the, actual / accurate / exact, reading (of blood glucose concentration) ; 2 re-usable ; 3 more precise reading / quantitative ;	1